On the Linkage Between the Asian Summer Monsoon and Tropopause Folds

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Abstract.

This study uses a set of idealized aquaplanet model experiments to investigate the linkage between the Asian summer monsoon circulation and tropopause fold activity. It's found that folds tend to occur on the northwestern side of the upper-level anticyclone associated with the monsoon circulation and are generated due to intensified monsoon circulation and resulting intensified subsidence. In addition, the impact of tropopause folds on extreme rainfall events is also examined. It's found that while the likelihood of extreme rainfall is largely decreased at and upstream of the occurrence of folds, the likelihood is significantly increased downstream. This pattern of suppression of extreme rainfall upstream and promotion downstream also persists for about 1-2 days, which likely suggests a positive feedback between extreme rainfall downstream and tropopause folds. Finally, changes in tropopause folds with monsoon intensity are also discussed.

Keypoints:

- Tropopause folds during northern summer occur as a result of monsoon dynamics
- Tropopause folds are associated with decreased likelihood of extreme rainfall upstream while increased likelihood downstream
- Tropopause folds vary with monsoon intensity

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1. Introduction

Tropopause folds are important dynamical processes of troposphere-stratosphere coupling and are characterized by deformation of the tropopause and intrusion of stratospheric air into the troposphere (e.g., *Reed*, 1955; *Reed and Danielsen*, 1959; *Danielsen*, 1968; *Shapiro*, 1980, 1981; *Keyser and Shapiro*, 1986). It is well known that tropopause folds are primarily driven by upper-level frontogenesis and ageostrophic vertical motion (e.g., *Keyser and Shapiro*, 1986 and references therein; *Thorpe*, 1997; *Bourqui and Trepanier*, 2010; *Skerlak et al.*, 2015). As a result, tropopause folds play an important role in stratosphere-troposphere exchange of ozone and other chemical constituents (*Lamarque and Hess*, 1994; *Holton et al.*, 1995; *Sprenger et al.*, 2003; *Stohl et al.*, 2003 and references therein). Furthermore, because folds are associated with development of upper-level positive potential vorticity anomalies, they could possibly contribute to rapid cyclogenesis, deep convective storms and other severe weather events (e.g., *Uccellini*, 1990; *Browning and Reynolds*, 1994; *Browning and Golding*, 1995; *Thorpe*, 1997; *Griffiths et al.*, 2000; *Goering et al.*, 2001; *Wernli et al.*, 2002).

Previous studies on tropopause folds mostly focused on the extratropical folds (e.g., *Sprenger et al.*, 2003; *Rao and Kirkwood*, 2005) and much less on the summer folds in the subtropics. However, recent studies by *Tyrlis et al.* [2014] and *Skerlak et al.* [2015] examined the climatology of tropopause folds using the ERA-Interim reanalysis and found a "hot spot" of summertime fold activity over the eastern Mediterranean and the Middle East. The contribution from these folds in carrying ozone-rich stratospheric air to the observed tropospheric summer ozone pool over the eastern Mediterrean is also dis-

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cussed in many other studies (e.g., *Roelofs and Lelieveld*, 1997; *Kourtidis et al.*, 2002; *Lelieveld et al.*, 2002; *Roelofs et al.*, 2003). Recent study of *Yang et al.* [2016] calculated the isentropic stratosphere-troposphere exchange and found a noticeable peak in stratosphere-to-troposphere exchange of ozone in late spring and early summer in the Northern Hemisphere (NH), likely associated with monsoon circulation.

Tyrlis et al. [2014] further argued from empirical evidence that recent upward trend of the fold activity over the eastern Mediterranean and the Middle East is controlled by the advanced onset of the South Asian summer monsoon in recent years and suggested a possible linkage between the Asian summer monsoon and tropopause fold activity. However, the dynamical mechanism is not fully understood. Some previous studies suggested Rossby wave breaking as a possible mechanism for tropopause folds, and in particular, Rossby wave breaking may take a form of tropopause folds in the vertical direction (e.g., *Postel and Hitchman*, 1999; *Allen et al.*, 2009; *Skerlak et al.*, 2015). However, during NH summer, tropopause folds and Rossby wave breaking do not take place at the same geographical location. More specifically, while tropopause folds occur over the eastern Mediterrean and the Middle East (e.g., *Skerlak et al.*, 2015), Rossby wave breaking events maximize over the North Pacific and North Atlantic oceans (e.g., *Postel and Hitchman*, 1999; *Homeyer and Bowman*, 2013; *Kunz et al.*, 2015).

Therefore, the main questions we aim to address in this study are: 1. What is the linkage between the Asian summer monsoon and tropopause folds? 2. Can we simulate and understand tropopause folds using idealized numerical model experiments? 3. What is the linkage between tropopause folds and extreme weather events? 4. How do tropopause folds vary with Asian summer monsoon intensity? This paper is organized as follows. In section

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2, we describe the reanalysis dataset, idealized model experiments, and identification methods for tropopause folds and Rossby wave breaking. In section 3, we present the results in idealized model experiments and discuss the dynamical mechanism and impact of tropopause folds. Section 4 concludes the paper.

2. Data and Methods

2.1. ERA-Interim Reanalysis

For observations, we use 6-hourly data during 1979-2015 from the ERA-Interim reanalysis produced by the European Centre for Medium-Range Weather Forecasts (*Dee and Coauthors*, 2011). In particular, we use zonal wind, meridional wind and temperature on a 0.75° longitude by 0.75° latitude grid and hybrid model levels to calculate potential temperature and potential vorticity (PV), which are further used for the identification of tropopause folds (to be discussed later). 6-hourly PV at 200hPa is also used for the identification of Rossby wave breaking (to be discussed later).

2.2. Idealized Model Experiments

In addition to observations, we also perform a set of idealized numerical model experiments using the NCAR Community Atmosphere Model (CAM) version 5 aquaplanet configuration with prescribed sea surface temperatures (SSTs) (*Neale and Coauthors*, 2012). The model is integrated at about 0.9° longitude by 0.9° latitude with 30 model levels in the vertical and the experiment setup is similar to *Shaw* [2014] and *Wu and Shaw* [2016]. The control experiment (namely "CTRL" run) prescribes a zonally symmetric SST with maximum SST located at 10°N (see Fig. S1(a) in the Supporting Information). The perturbation experiment (namely "7.5K" run) is designed to mimic the NH subtropical

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land-ocean heating contrast associated with the summer monsoon and further adds a subtropical zonal wave-1 SST perturbation that is centered at 30°N and has an amplitude of 7.5K (see Fig. S1(b) in the Supporting Information). Similar responses were found when a subtropical zonal wave-2 SST perturbation was introduced (*Shaw*, 2014; *Wu and Shaw*, 2016) and this study focuses on the zonal wave-1 perturbation experiment for the sake of simplicity. In addition, in order to examine the sensitivity of the results to the amplitude of the SST perturbation, two additional perturbation experiments are also performed with a SST perturbation amplitude of 5K and 10K, respectively (namely "5K" run and "10K" run). The CTRL, 5K and 10K simulations are integrated for 10 years. Since most of the results to be presented focus on the 7.5K run, this simulation is integrated for 20 years for the robustness of the results. All the analysis is based on the entire length of the simulations with the first 2 years of "spin-up" skipped, and we save 6-hourly zonal wind, meridional wind, temperature, specific humidity, precipitation and surface pressure.

Consistent with previous studies (e.g., *Shaw*, 2014; *Wu and Shaw*, 2016), as shown in Fig. 1, the 7.5K experiment is able to simulate a monsoon-like circulation with a low-level cyclonic circulation coupled to a strong ascent right above. A descent is seen on the northwestern side of the ascent, as a result of westward Rossby wave propagation, and is further coupled to an upper-level anticyclone (*Gill*, 1980; *Rodwell and Hoskins*, 1996, 2001).

2.3. Identification of Tropopause Folds

For the identification of tropopause folds, we make use of the algorithmn code that is developed by *Skerlak et al.* [2015]. The algorithmn code first identifies and labels the atmospheric air as five types including 1=troposphere, 2=stratosphere, 3=stratospheric

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cutoff or diabatically produced PV anomaly, 4=tropospheric cutoff, and 5=surface-bound PV anomaly based on the PV values and the 2PVU dynamical tropopause (or 380K tropopause, whichever is the lowest in altitude). Then tropopause folds are defined as multiple crossings of the dynamical tropopause in a vertical profile if, from highest to lowest altitude, there are subsequent transitions from label 2 to 1, then from 1 to 2, and finally from 2 to 1 (a schematic can be seen in Fig. 1 of *Skerlak et al.*, 2015).

Figure 2 shows an example of the identification of tropopause folds in the 7.5K experiment. The algorithmn outputs the pressure levels of multiple vertical crossings and the vertical extent of the tropopause folds.

2.4. Identification of Rossby Wave Breaking

For the identification of Rossby wave breaking (RWB), we use an algorithmn that is used in *Rivière* [2009] and *Lu et al.* [2014]. This RWB algorithmn is applied to 6-hourly PV at 200hPa and globe-circling contours are first identified. Then the algorithmn searches for overturning contours, and anticyclonic wave breaking (AWB) is identified when the contours overturn anticyclonically while cyclonic wave breaking (CWB) is found when the contours overturn cyclonically, similar to other algorithmns used in previous studies (e.g., *Strong and Magnusdottir*, 2008; *Homeyer and Bowman*, 2013).

3. Results

3.1. Simulated Folds in Idealized Experiments

First of all, Fig. 3(a) shows the observed frequency of tropopause folds during northern summer using the ERA-Interim reanalysis data averaged over 1979-2015. This is similar to Fig. 5 of *Skerlak et al.* [2015] and during northern summer, maximum frequency of

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tropopause folds occurs over the Mediterranean, Middle East and north Tibetan Plateau (about 10-20%, or equivalently 9-18 days on average during boreal summer) and to a lesser extent, western North America (about 2.5%, or about 2-3 days on average). Figure 3(a) also shows the upper-level horizontal circulation and as can be seen, tropopause folds tend to take place on the northwestern side of the upper-level anticyclone.

In the CTRL run with zonally symmetric SSTs, there is no occurrence of tropopause folds (not shown). Figure 3(b) shows the results in the idealized aquaplanet 7.5K experiment as a result of zonally asymmetric heating that mimics the contrast between warm land and cold ocean in summer. The figure is shifted by 120° longitude to the west as compared to Fig. 1 in order to better compare with the observational results in Fig. 3(a). To leading order, the simulated folds are similar to those in observations in both location and magnitude: simulated folds also tend to happen on the northwestern flank of the upper-level anticyclone with maximum frequency about 10-15%. But differences can be seen between the observed and simulated folds. For example, the simulation underestimates the tropopause fold frequency as compared to observations, and additionally, the idealized aquaplanet experiment produces a single maximum in tropopause fold frequency while the observations show two splitted maxima. The two splitted maxima of tropopause fold occurrence in reanalysis could likely be due to the two modes of variability of the South Asia High, i.e. the Tibetan Mode and the Iranian Mode (e.g., *Zhang et al.*, 2002) which are absent in idealized aquaplanet model experiments. In fact, the locations of maximum fold occurrence correspond well with the two modes shown in Fig. 3 of *Zhang et al.* [2002]. In summary, the idealized aquaplanet model experiments are able to capture the key features of the tropopause folds as in observations and both observations

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and idealized experiments show that there is a preferred location for the occurrence of the tropopause folds during northern summer.

3.2. Dynamical Mechanisms and Relationship with Rossby Wave Breaking

Next with the idealized aquaplanet simulations, we examine the dynamical mechanism underlying the occurrence of tropopause folds during northern summer. First, we make a composite of tropopause folding events and examine the surrounding circulation anomaly by removing climatological mean circulation in the 7.5K run in the neighborhood of 40° longitude and 20° latitude. As shown in Fig. 4(a)(b), tropopause folding activity is typically associated with an anomalously intensifed descent, which is consistent with previous studies on tropopause folds (e.g., *Keyser and Shapiro*, 1986 and references therein; *Thorpe*, 1997; *Bourqui and Trepanier*, 2010; *Skerlak et al.*, 2015). This descent anomaly is strongly coupled to a southward flow anomaly right below and an intensified low-level cyclonic circulation and ascent downstream (Fig. 4(b)) as well as a northward flow anomaly right above and an intensified upper-level anticyclonic circulation (Fig. 4(a)). A dipole of PV anomaly is also seen in the upper troposphere lower stratosphere with a positive PV anomaly in the north that is likely of stratospheric origin (Fig. 4(a)). Similar results are also found in the ERA-Interim reanalysis (not shown). Therefore, these results suggest that tropopause folds occur as a result of intensified monsoon circulation and resulting subsidence on the northwestern side.

As discussed in the Introduction, some studies have linked the occurrence of tropopause folds in the vertical direction to RWB in the horizontal direction (e.g., *Postel and Hitchman*, 1999; *Allen et al.*, 2009; *Skerlak et al.*, 2015). However, here we show that tropopause folds and RWB take place in different geographical locations during northern summer and

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both might likely occur as a result of monsoon dynamics. First, in reanalysis datasets, maximum AWB is found to occur over the eastern North Pacific ocean and eastern Europe and do not overlap with maximum occurrence of tropopause folds (see Fig. S2 in the Supporting Information, and the results are consistent with Fig. 9(a) of *Homeyer and Bowman* [2013] for example). Figure 5(a) shows the frequency of AWB and tropopause folds in the idealized aquaplanet experiment. Similar results are also found in idealized experiment in which AWB does not nearly overlap with tropopause folds but instead occurs downstream (i.e. 180-360°E, 30-50°N), and to a lesser extent, upstream of the upper-level anticyclone (i.e. 0-60°E, 40-50°N). CWB is not discussed here: CWB takes place at higher latitudes during northern summer in observations and occurs much less frequent than AWB in idealized experiments (not shown). It's also noted that the maximum frequency of folds is about 15% but the maximum frequency of AWB is about 5%. In addition, to illustrate how AWB varies with monsoon circulation, Fig. 5(b) shows the regression of AWB and tropopause folds on the monsoonal rising motion. It can be seen that, as monsoon circulation becomes intensified with stronger ascent, tropopause folds become more likely, due to the compensating stronger descent, and also shift slightly to the west. For the AWB upstream of the upper-level anticyclone, the frequency decreases strongly and shifts slightly to the west. And for the AWB downstream, there is generally a westward shift and waves tend to break in closer proximity to the upper-level anticyclone. Although the investigation of RWB dynamics is beyond the scope of this study, the regression results suggest that the monsoon circulation plays an important role in controlling the occurrence of both the tropopause folds and RWB.

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Therefore, both tropopause folds and RWB likely occur as a result of monsoon dynamics and this is summarized by a schematic in Fig. 5(c). While tropopause folds occur in conjunction with strong descent and strong upper-level northeastward flow, AWB takes place both downstream and upstream of the upper-level anticyclone associated with southwestward flow.

3.3. Impact on Extreme Rainfall Events

As discussed in the Introduction, previous studies have suggested the possible contribution of tropopause folds to rapid cyclogenesis and thus severe weather events. On the other hand, the subsidence associated with folds, as discussed in previous section, and also resulting dry air intrusion from the stratosphere seem to imply that it's unlikely that folds can instigate severe weather events (e.g., *Mapes and Zuidema*, 1996; *Fadnavis and Chattopadhyay*, 2017). Therefore, here we examine the impact of tropopause folds on extreme rainfall events using idealized model experiments.

In particular, we define a ratio $FR = \frac{P(\text{PRECT}\geq\text{threshold}|\text{folds})}{P(\text{PRECT}\geq\text{threshold})}$, which compares the likelihood of extreme rainfall events given the occurrence of tropopause folds to that under normal conditions. The existence of folds increases (or decreases) the likelihood of extreme rainfall as compared to normal conditions if FR is larger (or smaller) than 1. Here the threshold of extreme rainfall is set to be 95th percentile, as an example (similar results are found for other thresholds; not shown). Figure 6 shows the impact of tropopause folds on extreme rainfall before, during and after the occurrence of folds. During folds, the frequency of extreme rainfall is significantly reduced (about halving the likelihood) at and upstream of the folds, likely due to the descending motion associated with the folds. However, the likelihood of extreme rainfall is largely increased (almost doubling

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the likelihood) downstream of the folds likely associated with the ascent downstream. Previous studies also found similar responses in convection, and for example, *Allen et al.* [2009] found convection inhibition in the dry pool itself associated with tropopause folds but convection promotion at the leading edge of the dry slot in case studies from observations. And in our idealized model experiment, this suppression upstream and promotion downstream is seen to persist for at least 1-2 days after the occurrence of folds.

Figure 7 further shows the 6-hourly evolution of the maximum and minimum of FR. It is interesting to note that both the maximum and minimum of the likelihood are reached 6 hours after the occurrence of folds. This result is statistically significant at the 95% confidence level (not shown) and suggests a positive feedback between extreme ascent and rainfall downstream and tropopause folds. It is well known that an upper-level positive PV anomaly and a lower-level positive PV anomaly tend to phase lock, mutually intensify and promote cyclogenesis, which is likely what happens here between tropopause folds and extreme rainfall events (e.g., *Bretherton*, 1966; *Hoskins et al.*, 1985; *Hoskins and Berrisford*, 1988; *Bluestein*, 1993; a schematic of cyclogenesis associated with an upper-level PV anomaly can be seen in Fig. 21 of *Hoskins et al.*, 1985). An intensified low-level cyclonic circulation associated with the monsoon circulation can generate a strong descent on the northwestern side, which results in tropopause folds. Positive PV anomaly is then advected from the stratosphere during folds and causes the isentropic surfaces to bow towards it, leading to increased static stability towards positive PV anomaly while decreased static stability both above and below and thus further intensifies the ascending motion and rainfall. Figure 8 shows the evolution of potential temperature surfaces at about 5°S relative latitude where maximum extreme rainfall takes place. As can be seen,

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the isentropic surfaces in the lower troposphere are tilted downstream of the folds, which suggests decreased static stability and is consistent with increased frequency of extreme rainfall in the region. It is noted that the lower tropospheric isentropic surfaces are the most tilted during, 6 hours and 12 hours after the occurrence of folds. Similar results are found at around 0°N relative latitude (see Fig. S3 in the Supporting Information).

Although the occurrence of tropopause folds likely increases the probability of extreme rainfall downstream, the circulation system is strongly coupled between the lower and upper troposphere and the explicit role of folds alone on lower tropospheric circulation would need additional analysis such as PV inversion (e.g., *Griffiths et al.*, 2000; *Coronel et al.*, 2015). Such PV inversion analysis was done, for instance, by *Coronel et al.* [2015] where they examined the role of upper-level PV and moist processes in idealized surface cyclones and found a stronger surface cyclone with upper-level PV included than without.

Therefore, in summary, the idealized model experiments reveal that the occurrence of tropopause folds significantly decreases the likelihood of extreme rainfall events both at and upstream of the folds while it significantly increases the likelihood downstream. The results also suggest a positive feedback between extreme rainfall and tropopause folds, i.e. while tropopause folds occur as a result of intensified monsoon circulation, the resulting upper-tropospheric positive PV anomaly can modify the isentropic surfaces, decrease the static stability in the lower troposphere and further promote the ascending motion and rainfall.

3.4. Response to Monsoon Intensity

Finally, we use idealized aquaplanet model experiments to show the sensitivity of the tropopause folds to monsoon intensity. Figure 9 shows the simulated fold frequency in 5K,

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7.5K and 10K experiments. As can be seen, fold frequency increases significantly from 5K to 7.5K run, accompanied by significant intensified subsidence. From 7.5K to 10K run, folds have about the same magnitude of frequency but shift westward, which follows the westward shift of the subsidence. The change in tropopause fold frequency in response to anthropogenic forcing and changing monsoon intensity has a lot of implications for the stratosphere-troposphere transport of ozone and other chemical constituents and will be further studied in future work.

4. Conclusions and Discussion

Using a set of idealized aquaplanet model experiments forced with localized surface heating, we have demonstrated that tropopause folds occur because of monsoon dynamics during northern summer. More specifically, tropopause folds tend to occur on the northwestern side of the upper-level anticyclone associated with large-scale subsidence. Folds are typically seen to take place when there is an intensified subsidence associated with an intensified monsoon circulation. We also note that tropopause folds and AWB do not happen in the same geographical location and while tropopause folds occur on the northwestern side of the upper-level anticyclone, AWB takes place downstream and to a lesser extent, upstream. In addition, we have also quantified the impact of tropopause folds on extreme precipitation and we summarize the key conclusions in a schematic in Fig. 10. As shown in Fig. 10, we have found that while the likelihood of extreme rainfall is largely decreased at and upstream of the folds, the likelihood is significantly increased downstream. This pattern of suppression of extreme rainfall upstream while promotion downstream is seen to persist for 1-2 days, suggesting that a positive feedback between ascent downstream and tropopause folds is likely to play a role. As intensified monsoon

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circulation causes tropopause folds, the resulting positive PV anomaly associated with folds can lead to a decrease of static stability in the lower troposphere and further promote ascent and rainfall. Finally, tropopause folds also vary with monsoon intensity, i.e. the frequency of folds significantly increases from weak to medium monsoon intensity and then saturates from medium to strong monsoon intensity with a shift in location towards the west.

While the aquaplanet model experiments provide significant insights into the mechanisms responsible for the occurrence of tropopause folds and its impact on extreme weather events, there are a couple of caveats worth mentioning. First, convection in CAM5, or state-of-the-art atmospheric general circulation models in general, is highly parameterized, and frontal systems that are potentially important for folds (e.g., *Keyser and Shapiro*, 1986; *Rao and Kirkwood*, 2005) are not explicitly resolved. This study instead presents a picture of the interactions between large-scale circulation patterns and tropopause folds. Second, although the occurrence of tropopause folds likely increases the probability of extreme rainfall downstream, it's important to keep in mind that the lower and upper tropospheric circulation are strongly coupled together and it's not easy to untangle the two. However, previous studies have used PV inversion to examine the role of upper-level PV on surface cyclogenesis and found significant intensification of surface cyclones due to upper-level positive PV (e.g., *Griffiths et al.*, 2000; *Coronel et al.*, 2015).

The results highlight a strong linkage between the Asian summer monsoon circulation and tropopause fold activity over the eastern Mediterranean and the Middle East and suggest changes in fold frequency with changing monsoon intensity. Therefore, future work will be devoted to investigating the future changes in tropopause folds and resulting

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changes in stratosphere-to-troposphere transport of ozone and other chemical constituents as a consequence of anthropogenic forcing and changing monsoon intensity. Furthermore, the idealized simulations also demonstrate a potentially significant impact of tropopause folds on extreme rainfall events. The results highlight the need to consider upper troposphere lower stratosphere processes in studies of convection over the summer monsoon regions.

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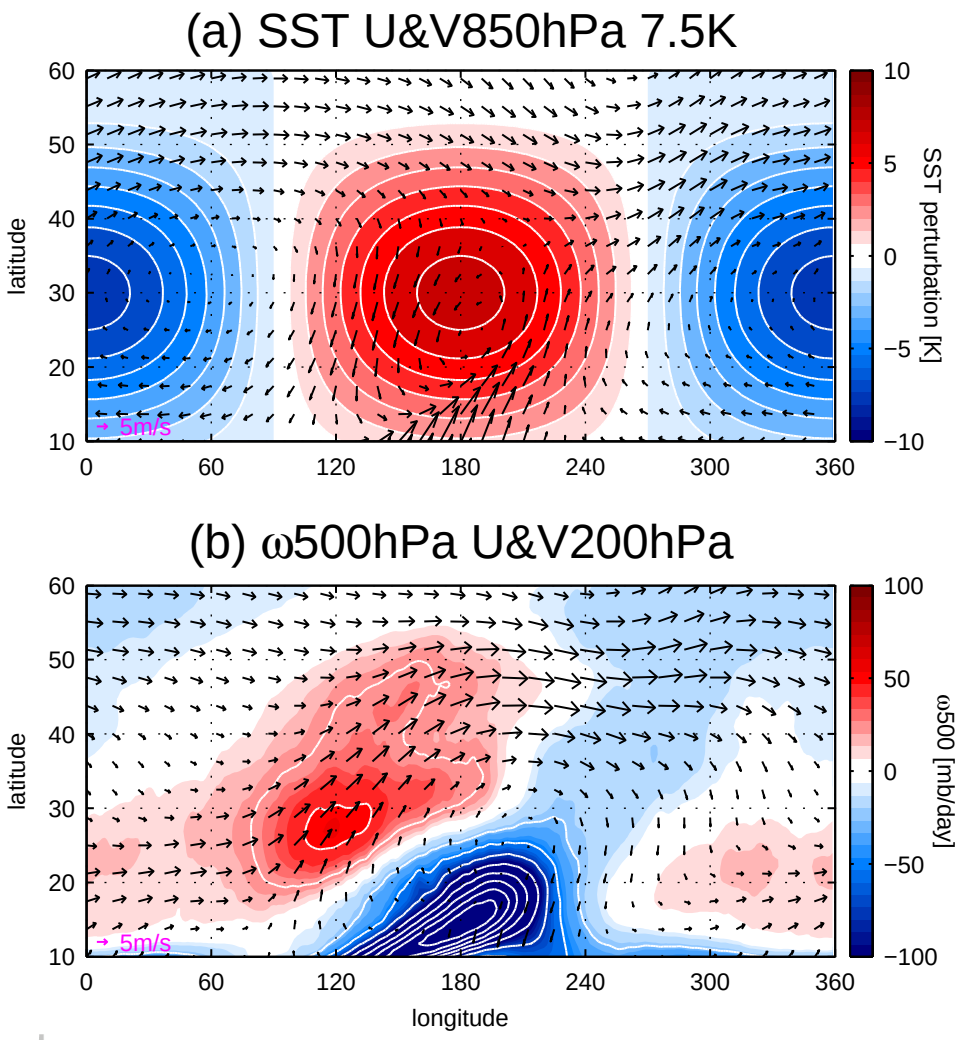

Figure 1. In idealized aquaplanet 7.5K model experiment, (a) zonal wave-1 SST perturbation that is maximized at 30°N with a maximum intensity of 7.5K (in unit of K, plotted in color shadings and contours with contour interval (CI) of 1K) and 850hPa horizontal velocity (vectors), and (b) 500hPa vertical velocity (in unit of mb/day, plotted in color shadings and contours with CI=25 mb/day) and 200hPa horizontal velocity (vectors).

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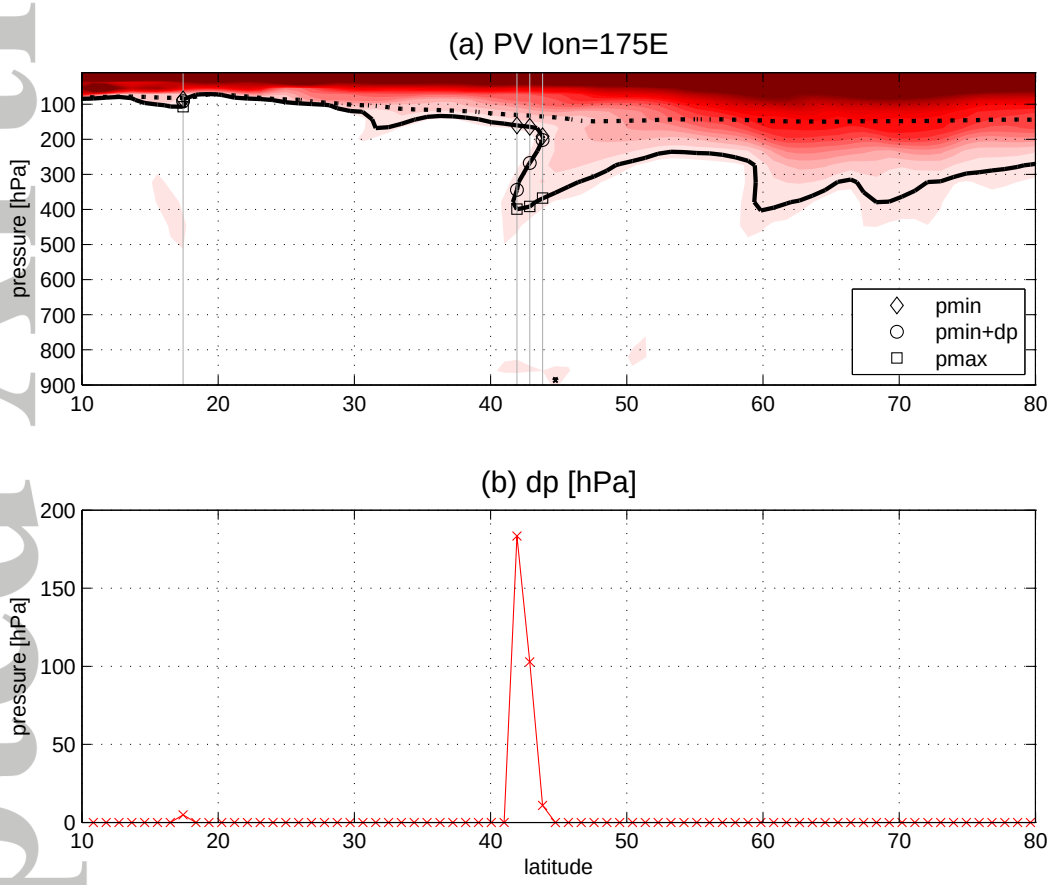

Figure 2. An example of tropopause fold in idealized aquaplanet 7.5K model experiment (model setup shown in Fig. 1) with (a) PV values (shown in color shadings) and (b) vertical extent of tropopause fold (dp) at 175E. Also shown in (a) are 2PVU surface (thick black line), 380K surface (black dotted line), and tropopause folds with multiple vertical crossings denoted by $p_{\min}$ (diamonds), $p_{\min} + dp$ (circles), and $p_{\max}$ (squares).

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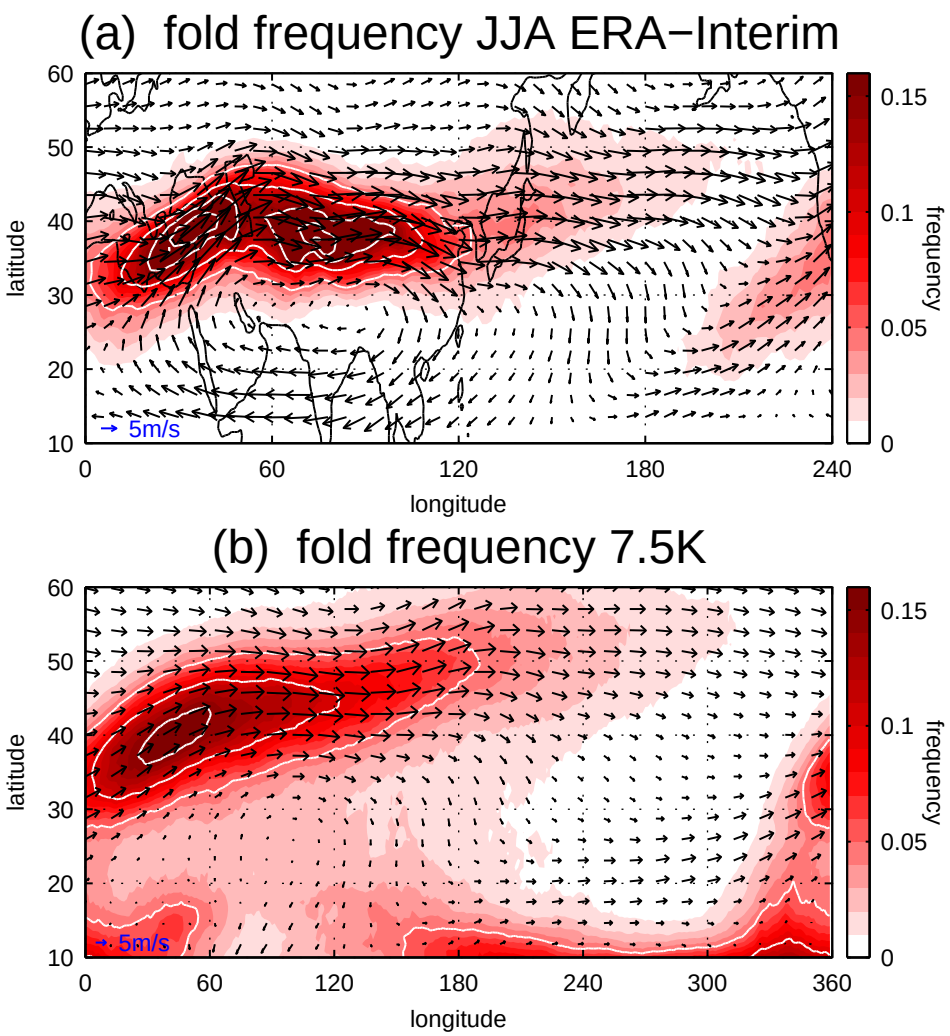

Figure 3. Tropopause fold frequency in (a) ERA-Interim reanalysis during June-July-August (JJA) and (b) idealized 7.5K model experiment (shown in color shadings and contours with CI=0.05). Also shown are 200hPa horizontal wind fields (vectors). The result from reanalysis is shown over 0°-240°E so that the size of the forcing domain and the fold response is better comparable to that of the idealized run. (b) is shifted by 120° longitude to the west as compared to Fig. 1 in order to better align with Fig. 3(a).

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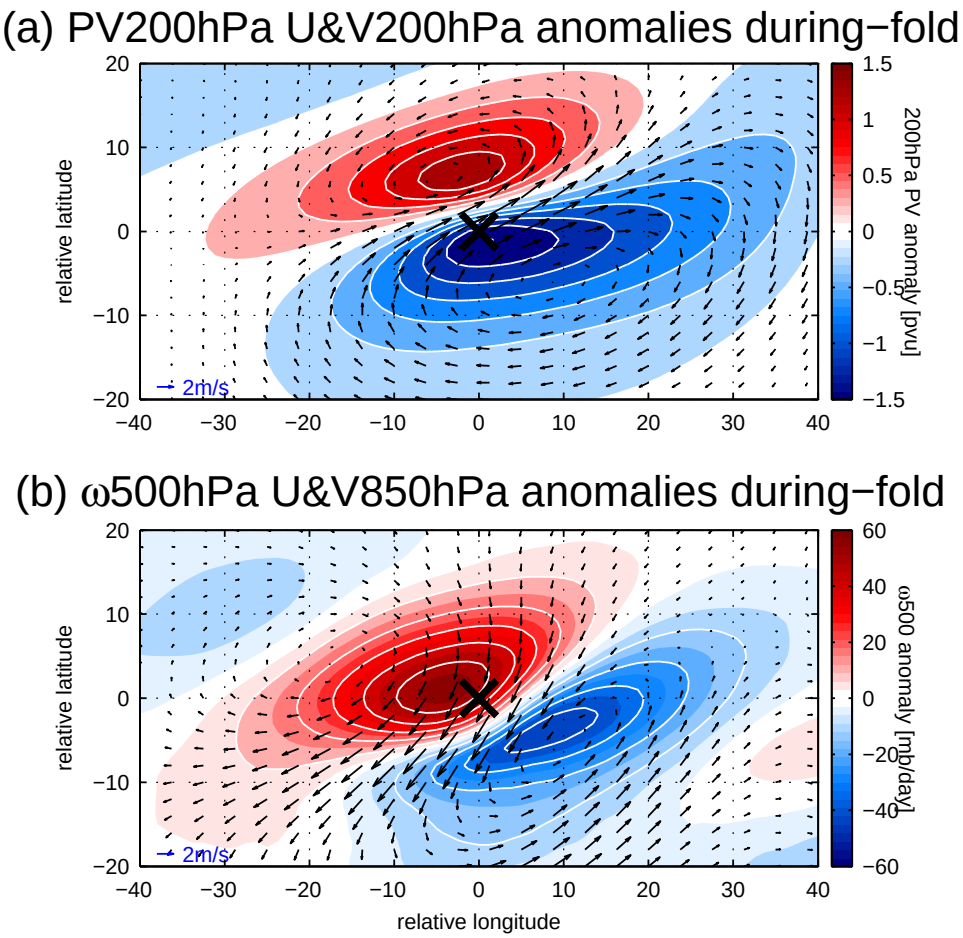

Figure 4. Composites of (a) 200hPa PV anomaly (in unit of PVU, plotted in color shadings and contours with CI=0.25PVU) and 200hPa horizontal velocity anomaly (vectors), and (b) 500hPa vertical velocity anomaly (in unit of mb/day, plotted in color shadings and contours with CI=10 mb/day) and 850hPa horizontal velocity anomaly (vectors) in idealized 7.5K run. Black crossing at the center highlights the tropopause fold location and both (a)(b) are shown as a function of relative longitude and relative latitude. The results are all statistically significant at the 95% confidence level.

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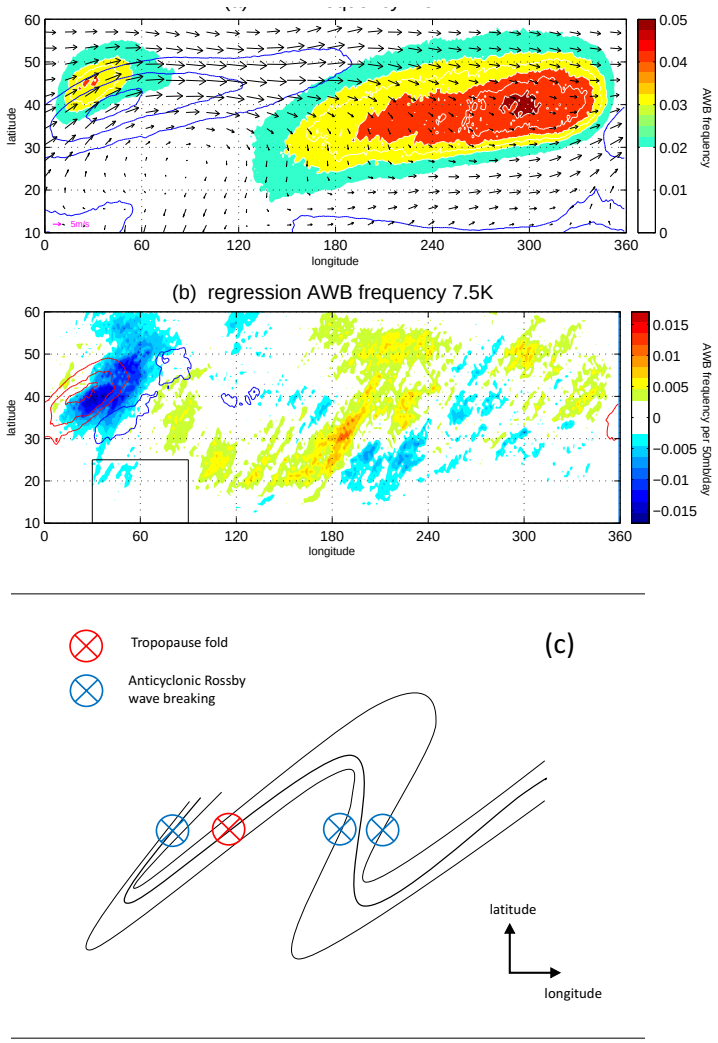

Figure 5. (a) Frequency of AWB (color shadings and white contours with CI=0.005) and tropopause folds (blue contours with CI=0.05) in idealized 7.5K aquaplanet experiment. Also plotted are climatological 200hPa horizontal velocity fields (vectors). (b) Regression of AWB and tropopause folds on 500hPa vertical velocity averaged over the black box (30-90°E and 10-25°N). The regression results for AWB are shown in color shadings and the unit is per 50 mb/day. The regression results for tropopause folds are shown in contours with CI=0.025 per 50 mb/day and red for positive values and blue for negative values. The regression results are all statistically significant at the 95% confidence level. (a)(b) are similar to Fig. 3(b) and are shifted by 120° longitude to the west as compared to Fig. 1. (c) Schematic showing the PV contours and occurrence of tropopause folds and AWB. Tropopause folds occur due to strong descent and strong northeastward flow while AWB takes place downstream and, to a lesser extent, upstream with southwestward flow.

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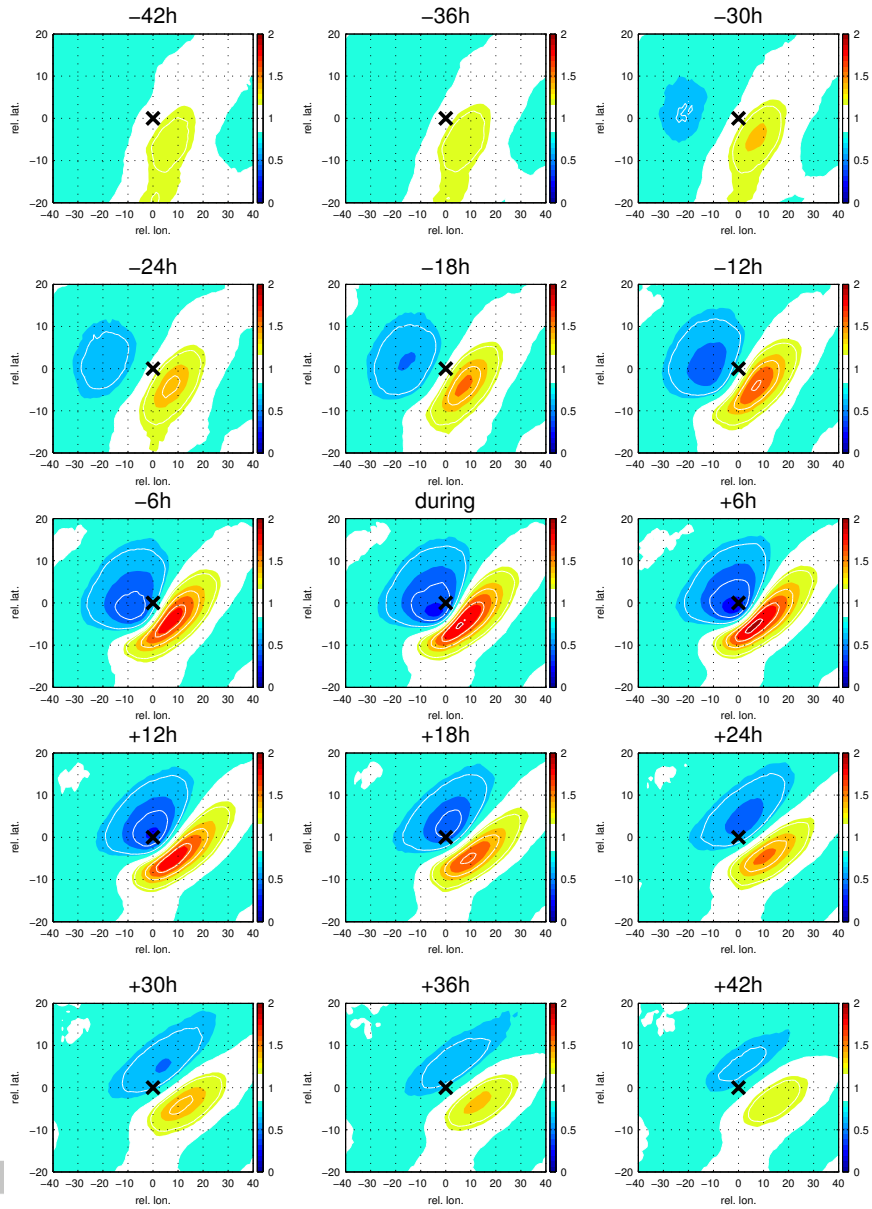

Figure 6. The impact of folds on extreme precipitation measured by $FR = \frac{P(\text{PRECT}\geq\text{threshold}|\text{folds})}{P(\text{PRECT}\geq\text{threshold})}$ where threshold is set to be 95th percentile here, shown in color shadings and white contours with CI=0.25 in 7.5K run. The results are shown every 6-hourly before, during and after the occurrence of tropopause folds. The results are all statistically significant at the 95% confidence level. Black crossings at the center highlight the tropopause fold location and the plots are shown as a function of relative longitude and relative latitude.

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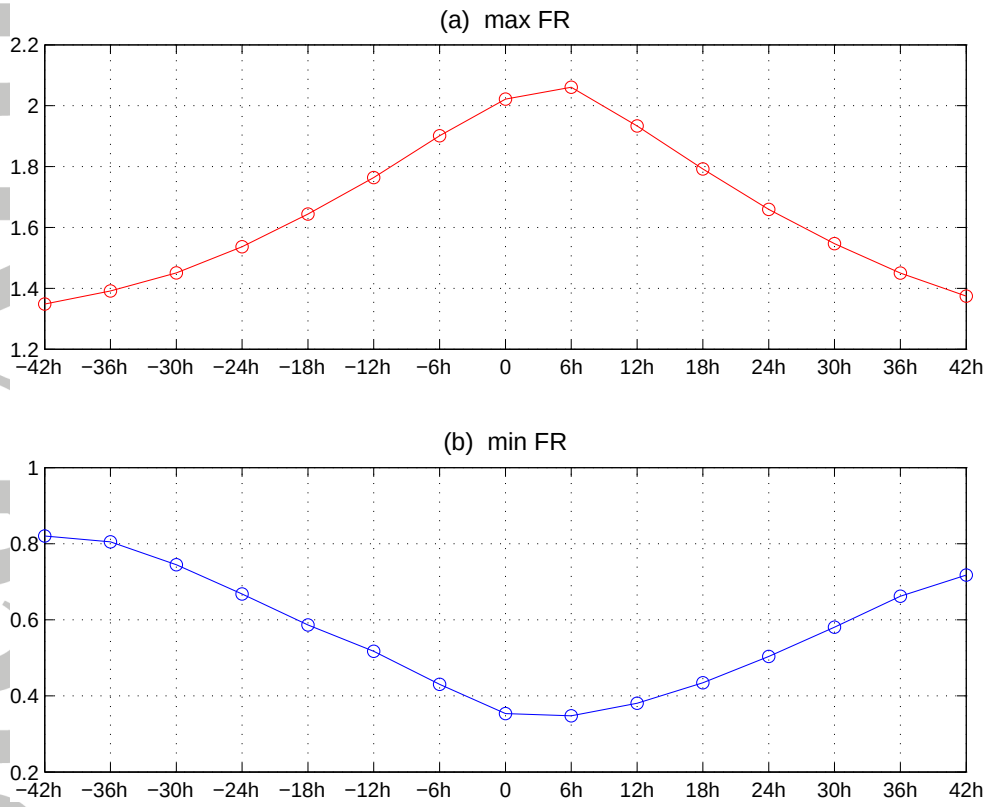

Figure 7. 6-hourly evolution of (a) maximum and (b) minimum of FR in 7.5K run. The results are all statistically significant at the 95% confidence level.

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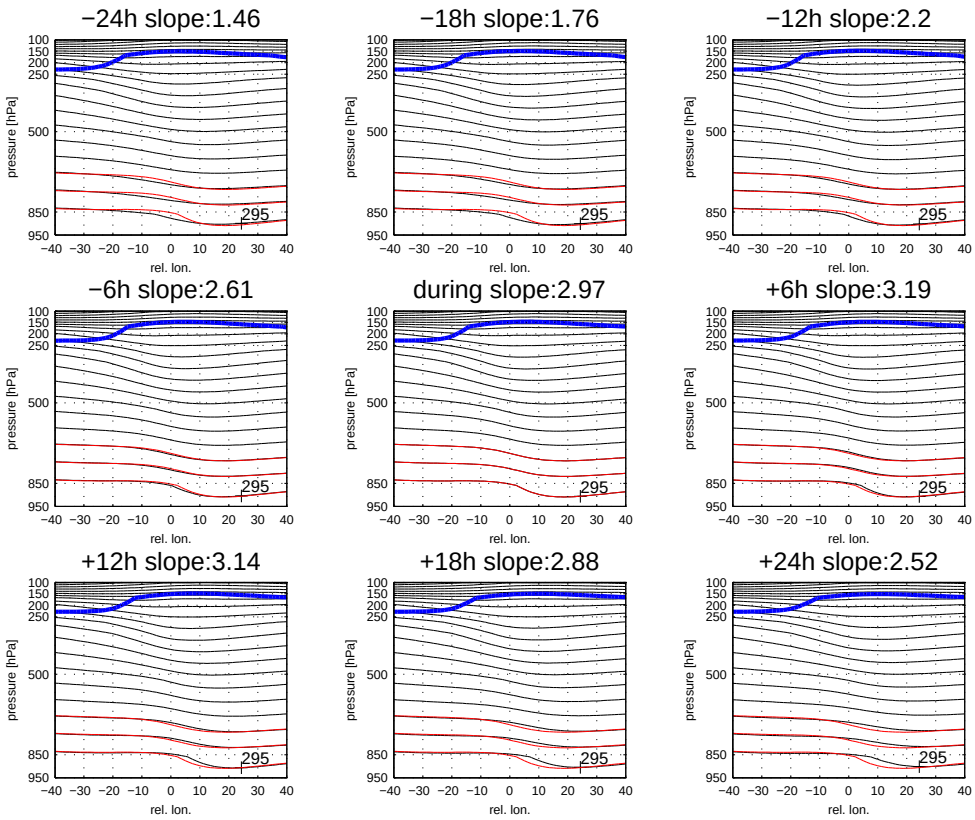

Figure 8. Cross section of potential temperature surfaces at relative latitude 5°S where maximum extreme rainfall occurs as a function of relative longitude and pressure level (black contours with CI=5K) every 6-hourly before, during and after the occurrence of tropopause folds in 7.5K run. The results focus on 1 day before and 1 day after the occurrence of folds. The numbers in the title indicate the isentropic slopes (in unit of hPa per degree longitude) at 295K surface and averaged downstream 0-20° relative longitude. Red contours are identical in all subplots and show the lower tropospheric isentropic surfaces during the occurrence of folds as a reference. Blue contours plot the 2PVU surfaces.

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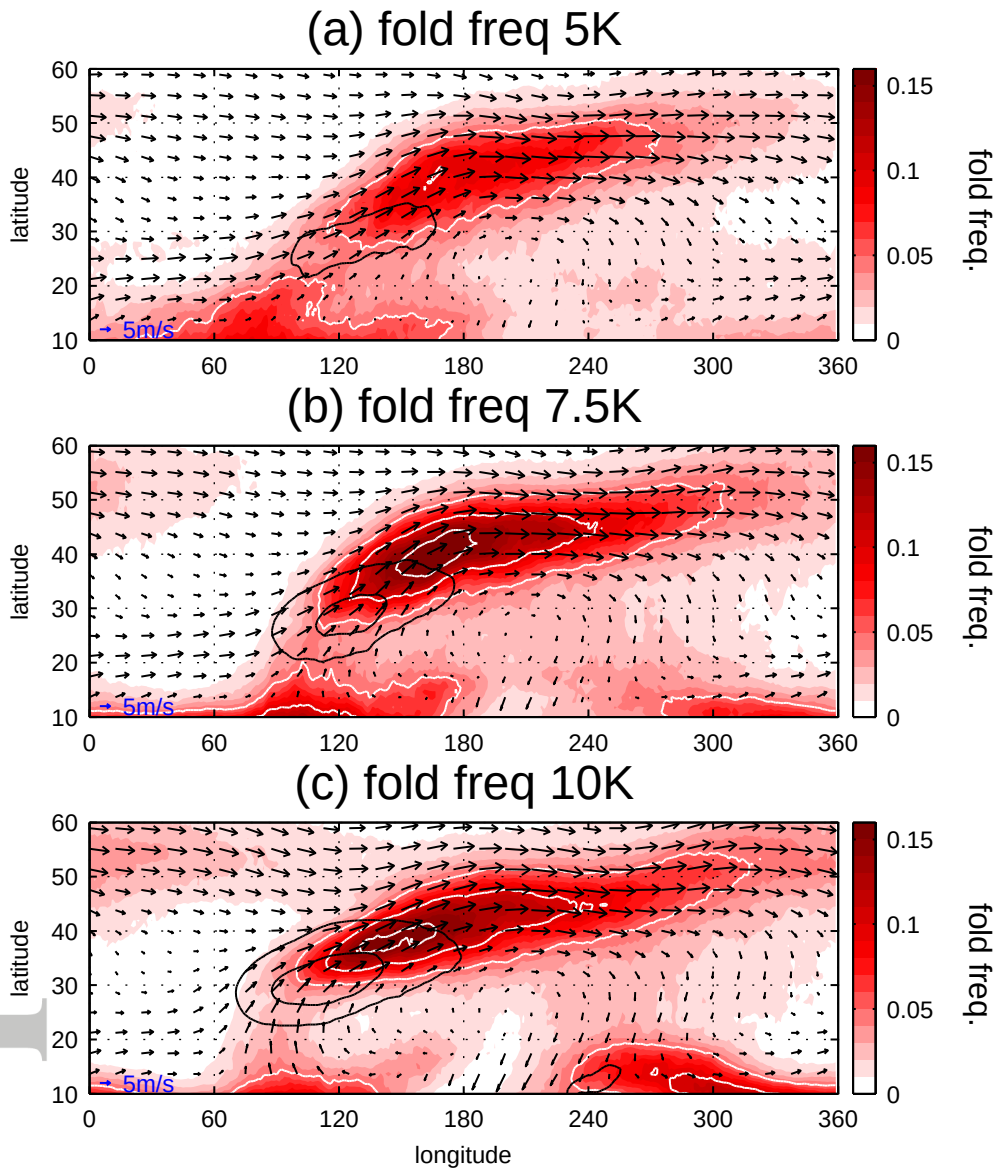

Figure 9. Tropopause fold frequency in (a) 5K, (b) 7.5K, and (c) 10K experiments, shown in color shadings and white contours with CI=0.05. Also plotted are 200hPa horizontal velocity fields (vectors) and 500hPa vertical velocity (in unit of mb/day, plotted in black contours with CI=25 mb/day).

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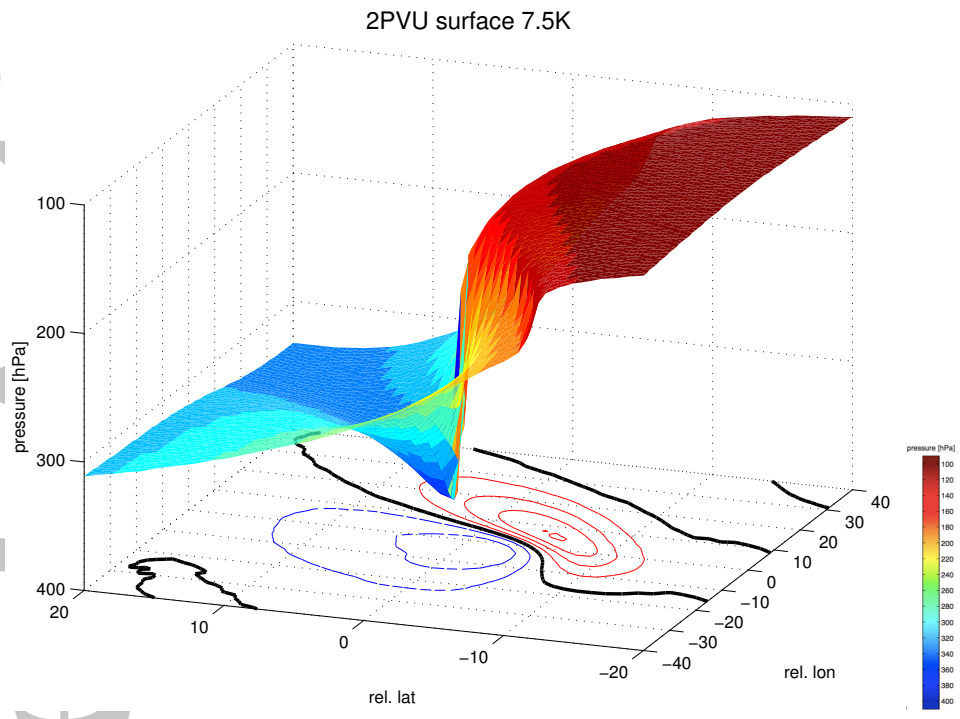

Figure 10. Similar to Fig. 6 but a three-dimensional view of 2PVU surface (color shadings) during the occurrence of tropopause folds. The contours plot the impact of folds on extreme precipitation measured by FR with blue contours for $FR < 1$, red for $FR > 1$ and thick black for $FR = 1$ with CI=0.25.

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